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PROCESSING SECTOR: XLE  
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--- Granger Causality: Macro -> XLE -

[NFP\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*  
Lag 2: p-value = 0.0011 \*\*\*  
Lag 3: p-value = 0.0198 \*\*  
Lag 4: p-value = 0.0289 \*\*  
Lag 5: p-value = 0.0756 \*  
Lag 6: p-value = 0.0765 \*  
Lag 7: p-value = 0.0277 \*\*  
Lag 8: p-value = 0.0496 \*\*  
Lag 9: p-value = 0.0425 \*\*

[Retail\_Sales\_Surprise]

Lag 1: p-value = 0.0003 \*\*\*  
Lag 2: p-value = 0.0408 \*\*  
Lag 3: p-value = 0.1909  
Lag 4: p-value = 0.4566  
Lag 5: p-value = 0.8010  
Lag 6: p-value = 0.8767  
Lag 7: p-value = 0.9076  
Lag 8: p-value = 0.9494  
Lag 9: p-value = 0.9638

|                          | XLK     |        |     | XLE     |        |     | XLF     |        |     | XLV  |        |     |
|--------------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|------|--------|-----|
|                          | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef | P-Val  | Sig |
| const                    | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0  | 0.0000 | *** |
| L1.NFP_Surprise          | 0.0000  | 0.5546 |     | 0.0000  | 0.7997 |     | -0.0000 | 0.0643 | *   | 0.0  | 0.0998 | *   |
| L1.Retail_Sales_Surprise | -0.0000 | 0.3371 |     | 0.0000  | 0.0001 | *** | 0.0000  | 0.7626 |     | -0.0 | 0.7725 |     |
| L2.Retail_Sales_Surprise | 0.0000  | 0.2555 |     | -0.0000 | 0.0765 | *   | 0.0000  | 0.2910 |     | 0.0  | 0.3413 |     |
| L3.NFP_Surprise          | -0.0000 | 0.2494 |     | 0.0000  | 0.2458 |     | 0.0000  | 0.5074 |     | -0.0 | 0.0506 | *   |
| L4.NFP_Surprise          | 0.0000  | 0.4477 |     | -0.0000 | 0.0091 | *** | -0.0000 | 0.3615 |     | 0.0  | 0.2129 |     |
| L6.NFP_Surprise          | 0.0000  | 0.1771 |     | -0.0000 | 0.0010 | *** | 0.0000  | 0.9118 |     | 0.0  | 0.6427 |     |
| L7.NFP_Surprise          | -0.0000 | 0.1483 |     | 0.0000  | 0.0206 | **  | -0.0000 | 0.3180 |     | -0.0 | 0.5781 |     |
| L9.NFP_Surprise          | 0.0000  | 0.0737 | *   | -0.0000 | 0.9346 |     | 0.0000  | 0.5373 |     | 0.0  | 0.2553 |     |

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PROCESSING SECTOR: XLF

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--- Granger Causality: Macro -> XLF

[Trade\_Balance\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0000 \*\*\*

Lag 4: p-value = 0.0009 \*\*\*

Lag 5: p-value = 0.0059 \*\*\*

Lag 6: p-value = 0.0978 \*

Lag 7: p-value = 0.2287

Lag 8: p-value = 0.3163

Lag 9: p-value = 0.0353 \*\*

[M1\_Money\_Surprise]

Lag 1: p-value = 0.0944 \*

Lag 2: p-value = 0.3656

Lag 3: p-value = 0.6334

Lag 4: p-value = 0.8290

Lag 5: p-value = 0.9323

Lag 6: p-value = 0.9851

Lag 7: p-value = 0.9958

Lag 8: p-value = 0.9987

Lag 9: p-value = 0.9846

|                           | XLK     |        |     | XLE     |        |     | XLF     |        |     | XLV  |        |     |
|---------------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|------|--------|-----|
|                           | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef | P-Val  | Sig |
| const                     | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0  | 0.0000 | *** |
| L3.Trade_Balance_Surprise | 0.0000  | 0.5582 |     | 0.0000  | 0.3742 |     | 0.0000  | 0.4827 |     | 0.0  | 0.0867 | *   |
| L8.Trade_Balance_Surprise | -0.0000 | 0.9623 |     | 0.0000  | 0.1966 |     | 0.0000  | 0.0141 | **  | 0.0  | 0.9320 |     |
| L8.M1_Money_Surprise      | 0.0000  | 0.2095 |     | 0.0000  | 0.0374 | **  | 0.0000  | 0.3525 |     | 0.0  | 0.0129 | **  |
| L9.Trade_Balance_Surprise | 0.0000  | 0.9301 |     | -0.0000 | 0.7397 |     | -0.0000 | 0.0018 | *** | -0.0 | 0.8284 |     |
| L9.M1_Money_Surprise      | -0.0000 | 0.0560 | *   | -0.0000 | 0.0644 | *   | -0.0000 | 0.1916 |     | -0.0 | 0.0009 | *** |

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PROCESSING SECTOR: XLK  
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--- Granger Causality: Macro -> XLK

[CPI\_Surprise]

Lag 1: p-value = 0.0208 \*\*  
Lag 2: p-value = 0.1754  
Lag 3: p-value = 0.4096  
Lag 4: p-value = 0.7065  
Lag 5: p-value = 0.8353  
Lag 6: p-value = 0.8698  
Lag 7: p-value = 0.8567  
Lag 8: p-value = 0.9381  
Lag 9: p-value = 0.9708

[PPI\_Surprise]

Lag 1: p-value = 0.0001 \*\*\*  
Lag 2: p-value = 0.0124 \*\*  
Lag 3: p-value = 0.0636 \*  
Lag 4: p-value = 0.2315  
Lag 5: p-value = 0.5303  
Lag 6: p-value = 0.7413  
Lag 7: p-value = 0.8005  
Lag 8: p-value = 0.9316  
Lag 9: p-value = 0.9688

[FedFunds\_Diff]

Lag 1: p-value = 0.0180 \*\*  
Lag 2: p-value = 0.0000 \*\*\*  
Lag 3: p-value = 0.0001 \*\*\*  
Lag 4: p-value = 0.0002 \*\*\*  
Lag 5: p-value = 0.0002 \*\*\*  
Lag 6: p-value = 0.0000 \*\*\*  
Lag 7: p-value = 0.0000 \*\*\*  
Lag 8: p-value = 0.0000 \*\*\*  
Lag 9: p-value = 0.0000 \*\*\*

--- Granger Causality: Macro -> XLE

[CPI\_Surprise]

Lag 1: p-value = 0.0119 \*\*  
Lag 2: p-value = 0.0226 \*\*  
Lag 3: p-value = 0.0870 \*  
Lag 4: p-value = 0.3390  
Lag 5: p-value = 0.5298  
Lag 6: p-value = 0.6772  
Lag 7: p-value = 0.7106  
Lag 8: p-value = 0.7927  
Lag 9: p-value = 0.7824

[PPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*  
Lag 2: p-value = 0.0001 \*\*\*  
Lag 3: p-value = 0.0009 \*\*\*  
Lag 4: p-value = 0.0235 \*\*  
Lag 5: p-value = 0.1494  
Lag 6: p-value = 0.2529  
Lag 7: p-value = 0.4824  
Lag 8: p-value = 0.6000  
Lag 9: p-value = 0.6441

[FedFunds\_Diff]

Lag 1: p-value = 0.9799  
Lag 2: p-value = 0.2405  
Lag 3: p-value = 0.5516  
Lag 4: p-value = 0.3752  
Lag 5: p-value = 0.0052 \*\*\*  
Lag 6: p-value = 0.0122 \*\*  
Lag 7: p-value = 0.0015 \*\*\*  
Lag 8: p-value = 0.0031 \*\*\*  
Lag 9: p-value = 0.0000 \*\*\*

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PROCESSING SECTOR: XLF  
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===== PROCESSING SECTOR: XLV  
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--- Granger Causality: Macro ->

[CPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*  
Lag 2: p-value = 0.0000 \*\*\*  
Lag 3: p-value = 0.0000 \*\*\*  
Lag 4: p-value = 0.0000 \*\*\*  
Lag 5: p-value = 0.0000 \*\*\*  
Lag 6: p-value = 0.0000 \*\*\*  
Lag 7: p-value = 0.0000 \*\*\*  
Lag 8: p-value = 0.0000 \*\*\*  
Lag 9: p-value = 0.0002 \*\*\*

[PPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*  
Lag 2: p-value = 0.0000 \*\*\*  
Lag 3: p-value = 0.0000 \*\*\*  
Lag 4: p-value = 0.0000 \*\*\*  
Lag 5: p-value = 0.0000 \*\*\*  
Lag 6: p-value = 0.0000 \*\*\*  
Lag 7: p-value = 0.0000 \*\*\*  
Lag 8: p-value = 0.0000 \*\*\*  
Lag 9: p-value = 0.0001 \*\*\*

[FedFunds\_Diff]

Lag 1: p-value = 0.3612  
Lag 2: p-value = 0.1929  
Lag 3: p-value = 0.0508 \*  
Lag 4: p-value = 0.4023  
Lag 5: p-value = 0.0912 \*  
Lag 6: p-value = 0.5176  
Lag 7: p-value = 0.5925  
Lag 8: p-value = 0.7046  
Lag 9: p-value = 0.4628

XLK

--- Granger Causality: Macro -> XLV

[CPI\_Surprise]

Lag 1: p-value = 0.0716 \*  
Lag 2: p-value = 0.4646  
Lag 3: p-value = 0.4953  
Lag 4: p-value = 0.6496  
Lag 5: p-value = 0.8512  
Lag 6: p-value = 0.9303  
Lag 7: p-value = 0.9551  
Lag 8: p-value = 0.9724  
Lag 9: p-value = 0.9869

[PPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*  
Lag 2: p-value = 0.0408 \*\*  
Lag 3: p-value = 0.0654 \*  
Lag 4: p-value = 0.1163  
Lag 5: p-value = 0.2996  
Lag 6: p-value = 0.4515  
Lag 7: p-value = 0.5969  
Lag 8: p-value = 0.8019  
Lag 9: p-value = 0.8664

[FedFunds\_Diff]

Lag 1: p-value = 0.1174  
Lag 2: p-value = 0.0009 \*\*\*  
Lag 3: p-value = 0.0055 \*\*\*  
Lag 4: p-value = 0.0136 \*\*  
Lag 5: p-value = 0.0001 \*\*\*  
Lag 6: p-value = 0.0003 \*\*\*  
Lag 7: p-value = 0.0000 \*\*\*  
Lag 8: p-value = 0.0000 \*\*\*  
Lag 9: p-value = 0.0000 \*\*\*

XLE

XLF

XLV

|                  | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig |
|------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|---------|--------|-----|
| const            | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0000  | 0.0000 | *** |
| L1.CPI_Surprise  | -0.0000 | 0.5584 |     | 0.0001  | 0.3696 |     | 0.0002  | 0.0137 | **  | -0.0000 | 0.8002 |     |
| L2.CPI_Surprise  | 0.0000  | 0.6435 |     | -0.0002 | 0.1205 |     | -0.0002 | 0.0986 | *   | 0.0000  | 0.3248 |     |
| L2.FedFunds_Diff | -0.0007 | 0.0001 | *** | -0.0003 | 0.2184 |     | -0.0001 | 0.6065 |     | -0.0003 | 0.0009 | *** |
| L5.FedFunds_Diff | -0.0002 | 0.1791 |     | -0.0010 | 0.0001 | *** | -0.0004 | 0.2087 |     | -0.0003 | 0.0002 | *** |
| L6.FedFunds_Diff | 0.0008  | 0.0000 | *** | 0.0001  | 0.7784 |     | 0.0002  | 0.5663 |     | 0.0000  | 0.8983 |     |
| L7.FedFunds_Diff | -0.0008 | 0.0000 | *** | -0.0006 | 0.0338 | **  | -0.0001 | 0.6909 |     | -0.0006 | 0.0000 | *** |
| L8.FedFunds_Diff | 0.0001  | 0.6189 |     | 0.0000  | 0.9107 |     | 0.0001  | 0.6408 |     | 0.0002  | 0.0348 | **  |
| L9.FedFunds_Diff | -0.0001 | 0.7427 |     | -0.0010 | 0.0002 | *** | -0.0003 | 0.3110 |     | -0.0003 | 0.0001 | *** |